

HNSS04 · Clinical Problems in Craniofacial Development — Chart

LO9: typical malformations, their causes and anatomy · LO10: normal vs abnormal development on ultrasound | **STEP 1** tags mark rows that go beyond the lecture

HoyaDocs
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Facial Clefts — two prominences that failed to fuse

MALFORMATION	WHO / CAUSES	EMBRYOLOGY: WHAT FAILED	WHERE	PRESENTATION	DETECTION	MANAGEMENT & KEY FACTS
Lateral cleft lip <i>the two M's that missed</i> Anterior cleft	~1/1,000 births, 80% male ; anticonvulsants, lithium	Maxillary + medial nasal prominences fail to fuse (wk 5–8)	Upper lip, lateral to the philtrum; may extend through the alveolar ridge (primary palate)	Visible lip cleft at birth, unilateral or bilateral	Prenatal ultrasound (detectable with a skilled technician); exam at birth	Surgery (only treatment). Often occurs with cleft palate, but a separate embryologic cause
Cleft secondary palate + cleft uvula <i>the shelves that didn't meet</i> Posterior cleft	Anticonvulsants (phenobarbital, lamotrigine), lithium	Palatine shelves (from maxillary prominences) fail to fuse with each other, the nasal septum and/or the primary palate (wk 6.5–7 on)	Roof of the mouth behind the incisive foramen + uvula	Feeding problems, nasal regurgitation; recurrent otitis media STEP 1	Oral exam at birth; internal, so harder to see on ultrasound	Surgery . Shelves must rise above the tongue to meet
Oblique facial cleft <i>maxillary meets the wrong neighbor</i> Lip + palate	—	Maxillary fails to merge with the lateral nasal prominence	From the upper lip toward the medial eye	Cleft lip + cleft palate. The lip cleft is secondary to the cleft palate (lateral nasal doesn't form lip)	Prenatal ultrasound; exam	Surgery . Classic trap: lateral nasal ≠ lip
Median cleft of upper lip <i>the midline split</i> Rare	—	The two medial nasal prominences fail to merge	Midline of the upper lip	Midline lip notch; may come with a bifid nose / incomplete median cleft	Exam; ultrasound	Surgery
Median cleft of lower lip <i>the split chin</i> Rare	—	The two mandibular prominences fail to merge	Midline of the lower lip / jaw	Midline lower-lip cleft	Exam	Surgery
Macrostomia <i>the too-wide mouth</i> NC migration defect	Neural crest migration defect	Maxillary + mandibular prominences fail to merge at the mouth corner	Corner of the mouth	Abnormally wide mouth. Know-exist: single nostril + microstomia (small mouth)	Exam	Surgery

Drugs linked to facial clefts

DRUG	WHAT THE LECTURE SAYS	STEP 1 CLASSIC DEFECT
Lamotrigine	Registry data: oral clefts 2.5 per 1,000 exposed vs 0.37 per 1,000 controls. "Women have to be told about" this	Oral clefts
Phenobarbital	Older studies: ↑ cleft palate, though cleft palate was a small share of all deformities (6.5% vs 1% in controls)	—
Valproic acid	↑ craniofacial disorders	Neural tube defects (spina bifida)
Lithium	↑ craniofacial disorders	Ebstein anomaly
Phenytoin	(not named in lecture)	Fetal hydantoin : cleft lip/palate, hypoplastic nails, midface hypoplasia

★ **USMLE high-yield: All facial clefts can be caused by improper neural crest migration/function** **EXAM EMPHASIS** — a guaranteed exam topic. **Cleft lip** = maxillary + medial nasal ("the two M's"). **Cleft palate** = palatine shelves. **Oblique** = maxillary + lateral nasal. Lip and palate have separate causes but often occur together; treatment is surgical. Anticonvulsant teratogens: **phenytoin** (cleft lip/palate), **valproate** (neural tube defects).

Neck & Thyroid Malformations

MALFORMATION	WHO / CAUSES	EMBRYOLOGY: WHAT FAILED	WHERE	PRESENTATION	DETECTION	MANAGEMENT & KEY FACTS
Thyroglossal duct cyst <i>the midline lump that moves with swallowing</i> Benign	Children, young adults (congenital)	Remnant of the thyroglossal duct , which should disappear (by apoptosis) after the thyroid descends from the foramen cecum	Midline , anywhere on the descent path: below the hyoid , base of tongue, near the thyroid cartilage	Painless midline neck mass that moves up with swallowing / tongue protrusion	Exam; histology: respiratory epithelium + thyroid follicles (HNSS16)	Generally not a major clinical problem to correct. Excision (Sistrunk procedure, including mid-hyoid) STEP 1
Thyroglossal fistula <i>the cyst with an exit</i> Benign	Congenital; can be present at birth	Thyroglossal duct remnant stays open to the skin by a canal	Midline neck	Draining midline opening	Exam	Excision
Lingual thyroid <i>the thyroid that never left home</i> Ectopic STEP 1	Congenital	Thyroid fails to descend from the foramen cecum	Base of the tongue (most common ectopic site)	Mass at the tongue base; dysphagia; may be hypothyroid	Exam; thyroid scan	Confirm other functioning thyroid exists before removing — it may be the only thyroid

★ **USMLE high-yield:** Midline + moves with swallowing = **thyroglossal duct cyst**. Contrast **branchial cleft cyst**: **lateral**, anterior to the SCM, doesn't move (persistent cervical sinus, HNSS03).

Neural Crest & Chromosomal Syndromes

MALFORMATION	WHO / CAUSES	EMBRYOLOGY: WHAT FAILED	WHERE	PRESENTATION	DETECTION	MANAGEMENT & KEY FACTS
Treacher Collins <i>the first arch that didn't fill in</i> Autosomal dominant	Inherited (AD); very high retinoic acid in mice mimics it	Tcof-1 mutation → NC from R2/R4 fails to migrate into arches 1–2 (~wk 2–3.5)	Cheeks, jaw, eyelids, ears	Malformed zygomatic bones, mandibular hypoplasia, lower eyelid colobomas, malformed external ears	Ultrasound: small jaw, abnormal ears	Surgical reconstruction; airway risk from the small jaw STEP 1
Down syndrome (trisomy 21) <i>the missing nasal bone</i> Chromosomal	Extra chromosome 21; ↑ maternal age STEP 1	Chromosomal (nondisjunction) STEP 1	Face (nasal bone, ears)	Characteristic facies (among other findings)	Ultrasound: absent nasal bone , abnormal ear position; ↑ nuchal translucency STEP 1	Quad screen: ↓ AFP, ↑ β-hCG, ↓ estriol, ↑ inhibin A STEP 1

★ **USMLE high-yield:** Treacher Collins = **first-arch neural crest** (Tcof-1). Down syndrome first-trimester markers: ↑ **nuchal translucency** + **absent/hypoplastic nasal bone**.

LO10 · Reading a Fetal Face Ultrasound

Normal vs abnormal

WHAT YOU SEE	WHAT TO LOOK FOR	WHAT IT MEANS	TRAP
Normal fetal face	Identify neurocranium, viscerocranium, frontal prominence, nose, fused maxillary process	Normal craniofacial development	Don't mistake the fetal hand or ribs for facial structures
Ear position	Ears relative to the eye and nose	Low-set / malpositioned ears → arch 1–2 or chromosomal problem	—
Nasal bone	Is the nasal bone present?	Absent nasal bone → Down syndrome (trisomy 21)	—
Facial clefts	Lip (and sometimes palate) defects	Cleft lip ± palate	Isolated cleft palate is internal and harder to see

★ **USMLE high-yield:** On a fetal face ultrasound: identify the neurocranium, viscerocranium, frontal prominence, nose and fused maxillary process; then check “**ears and nose**” — ear position and the nasal bone. **Absent nasal bone = trisomy 21 marker.**